



# Module 4

## Multiprotocol Configuration

# About This Module



This module focuses on enabling you to do the following:

- Demonstrate a multiprotocol configuration that supports SMB and NFS access
- Explain the permission mapping between SMB and NFS data

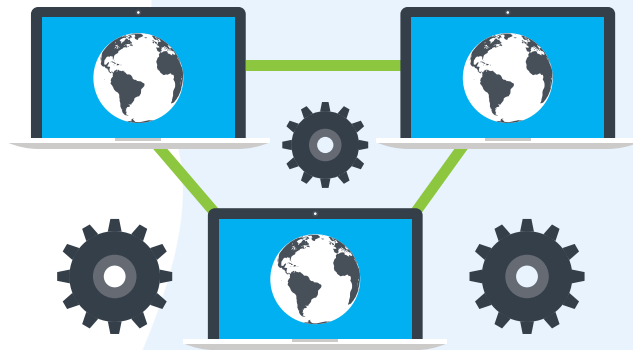


# Lesson 1

## Multiprotocol Configuration

# Why Multiprotocol?

- If SMB and NFS clients require access to the same files, a multiprotocol configuration is required.
- Windows and UNIX accounts do not explicitly match:
  - Windows users need access to files with UNIX file permissions.
  - UNIX users need access to files with NTFS permissions.
- Name mapping options:
  - Convert using regular expressions.
  - Map a default user account to all users that are not implicitly mapped or converted, using regular expressions.

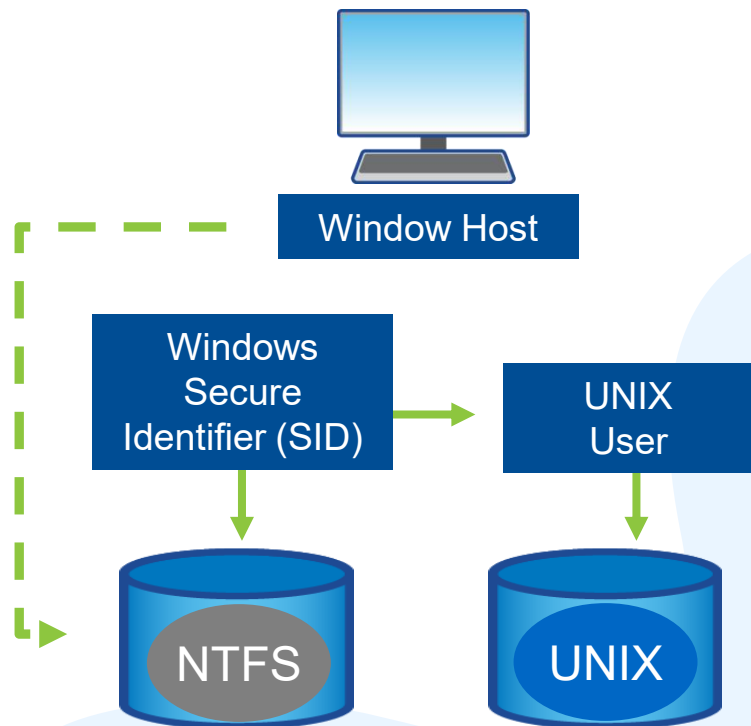


# Multiprotocol Access and Security Styles

- Volumes and qtrees can have the following permission styles:
  - NTFS security-style access control list (ACL)
  - UNIX style
  - Mixed
- The use of UNIX security-style permissions does not prevent Windows (SMB) users from accessing a volume or qtree, if multiprotocol is correctly configured.
- The use of NTFS security-style ACL permissions does not prevent UNIX (NFS) users from accessing a volume or qtree, if multiprotocol is correctly configured.

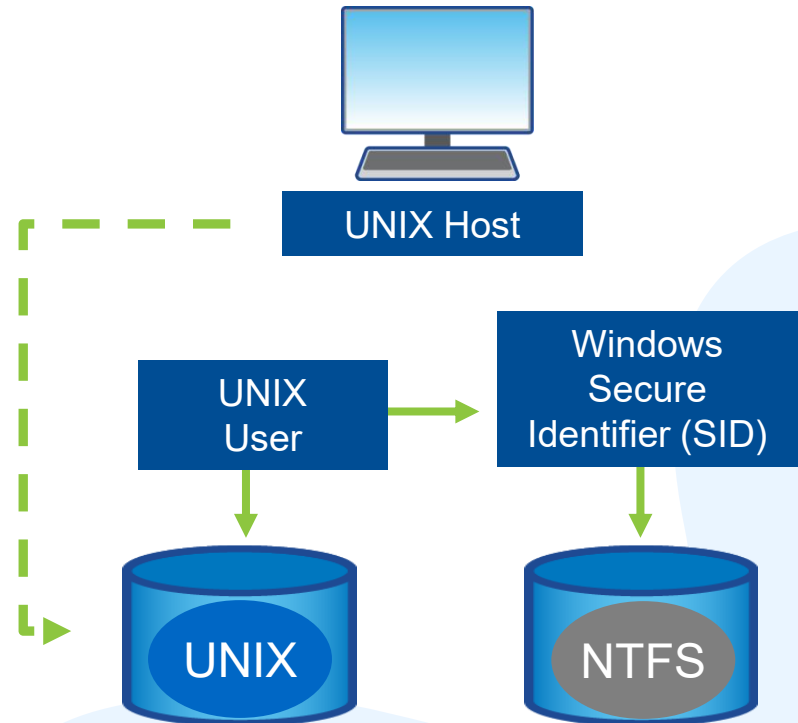
# Windows User Security Style Interaction

- To access an NTFS security-style volume or qtree, a Windows user is tested against NTFS security-style ACLs.
- To access a UNIX security-style volume or qtree, a Windows user must be mapped to a UNIX user ID (UID) and an associated UNIX group ID (GID).



# UNIX User Access to Files

- To access a UNIX security-style volume or qtree, a UNIX user is tested against the UNIX file permissions.
- To access an NTFS security-style volume or qtree, a UNIX user and group must be mapped to a Windows user and associated Windows groups.



# Security Styles and Effective Permissions

| Security Style | Clients that Can Modify Permissions | Permissions that Clients Can Use             | Resulting Effective Security Style | Clients that Can Access Files |
|----------------|-------------------------------------|----------------------------------------------|------------------------------------|-------------------------------|
| UNIX           | NFS                                 | NFSv3 mode bits<br>NFSv4.x ACLs              | UNIX                               | NFS and SMB                   |
| NTFS           | SMB                                 | NTFS ACLs                                    | NTFS                               | NFS and SMB                   |
| Mixed          | NFS or SMB                          | NFSv3 mode bits<br>NFSv4.x ACLs<br>NTFS ACLs | UNIX<br>UNIX<br>NTFS               | NFS and SMB                   |

NFSv3 = NFS Version 3  
NFSv4x = NFS Version 4.x

# Setting Security Styles

Changing a security style resets all security permissions within a volume or qtree to the default:

- NTFS: Everyone has read/write permission.
- UNIX: The user, group, and world have rwx (read write execute) permissions.

## ONTAP CLI

```
cluster1::>volume modify
               -vserver vsNFS
               -volume vol_NFS1
               -security-style unix
cluster1::>volume qtree security
               -vserver vsNFS
               -volume vol_NFS1
               -qtree qtree1
               -security-style unix
```

# Multiprotocol Configuration Workflow

- Storage virtual machine (SVM) preparation:
  - Create/configure the SVM with SMB, NFS.
  - Create/assign the SVM root volume export policy.
  - Configure DNS (domain name service) and assign the SMB server name to the data LIF IP.
  - Configure Lightweight Directory Access Protocol (LDAP) (if appropriate).
- Map UNIX and Windows usernames.
- Create the data volume and configure protocol access:
  - Create the SMB share and permissions.
  - Create and assign the export policy for NFS.
- Confirm that administrative access is properly configured:
  - Verify SMB access as a Windows administrator.
  - Verify NFS access from a UNIX administration host.
- Configure SMB and NFS clients and verify access.



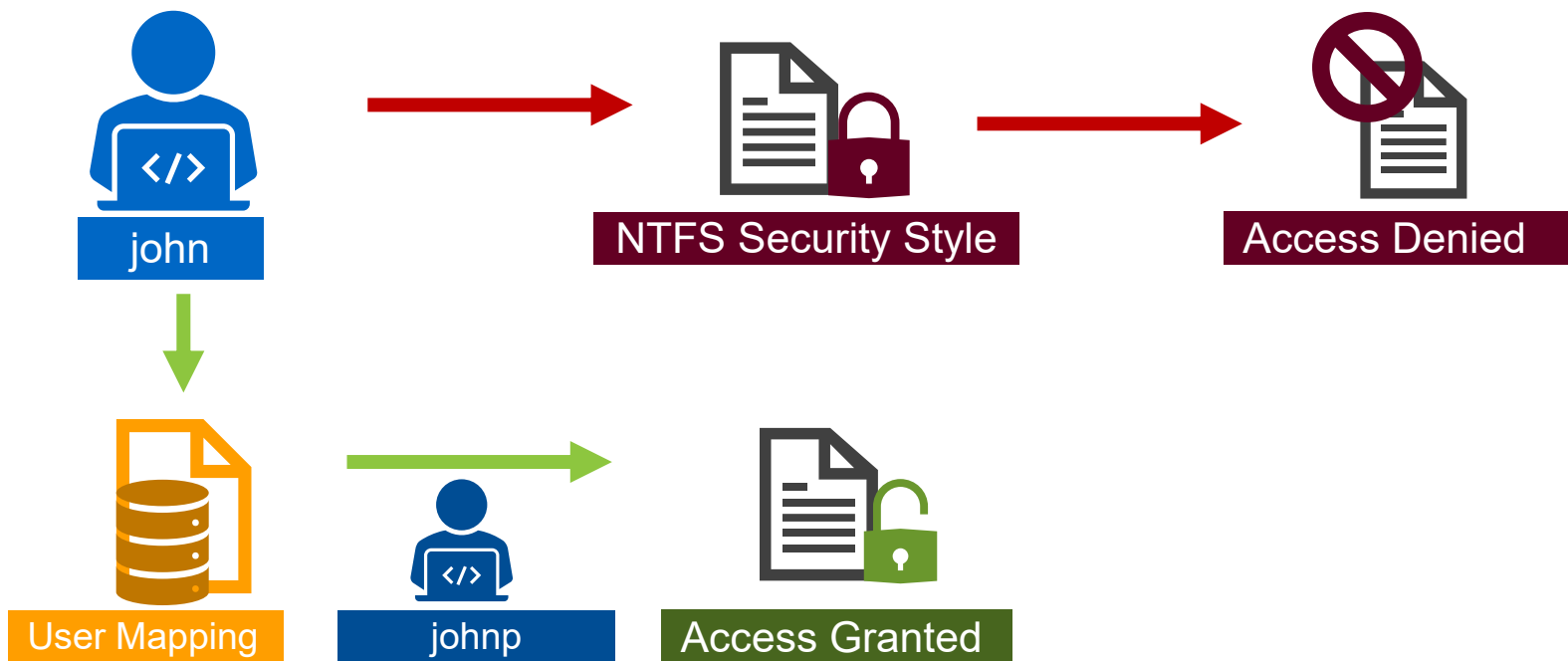
## Lesson 2

# Permission Mapping Between SMB and NFS

# The Need for Name Mapping

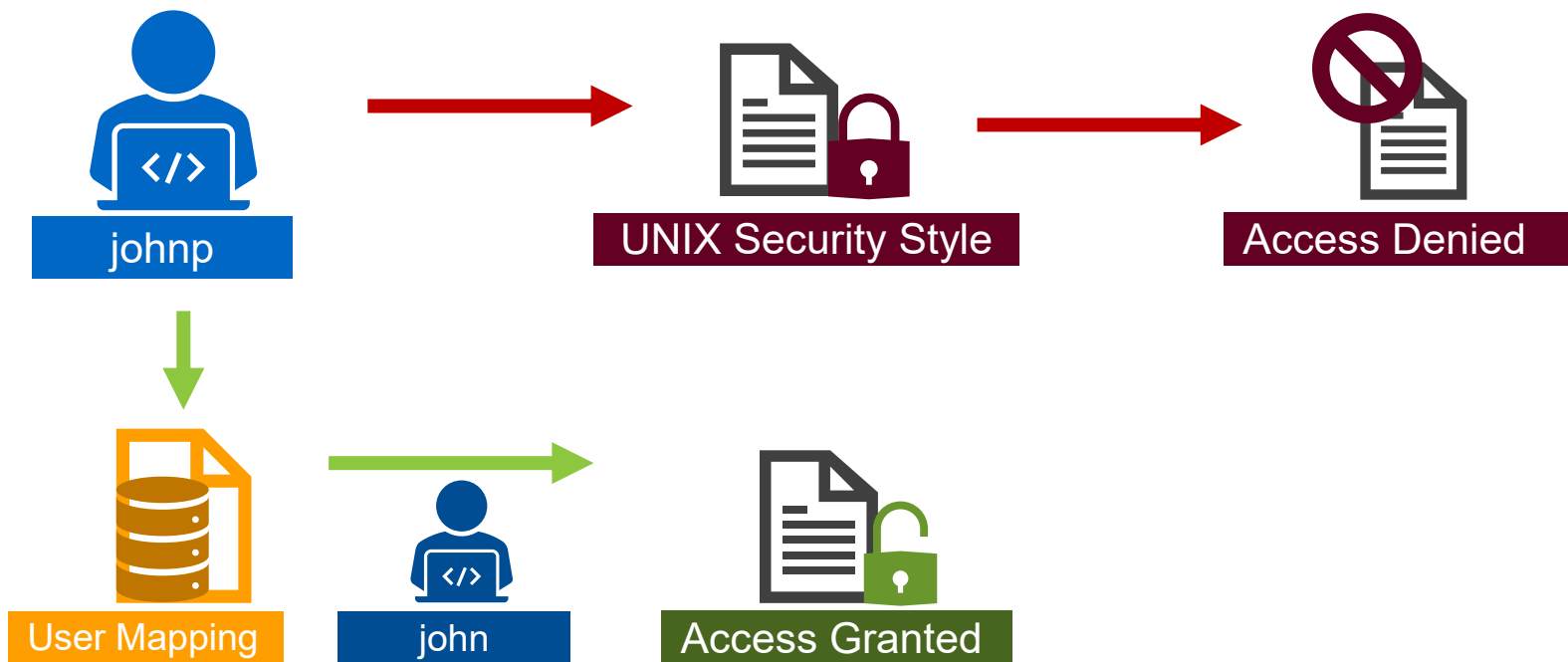
- ONTAP software uses name mapping to obtain user credentials to provide proper file access for NFS and SMB clients to the same files.
- Credentials must match the assigned UNIX or NTFS security style.
- ONTAP software uses name mapping to map the following:
  - SMB identities to UNIX identities
  - Kerberos identities to UNIX identities
  - UNIX identities to SMB identities
- Name mappings are not required in the following scenarios:
  - A pure UNIX environment
  - Use cases where all credentials are mapped to the same default user
- ONTAP supports up to 1,024 name mappings for each direction per SVM.

# Mapping a UNIX User to NTFS Permissions



```
cluster1::> vserver name-mapping create -vserver vs1 -direction unix-win  
-position 5 -pattern john -replacement "domainname\\johnp"
```

# Mapping a Windows User to UNIX Permissions



```
cluster1::>vserver name-mapping create -vserver vs1 -direction win-unix -  
position 1 -pattern "domainname\\johnp" -replacement "john"
```

# Name Mapping with Regular Expressions

- Windows-to-UNIX regular expressions are not case-sensitive.
- Kerberos-to-UNIX and UNIX-to-Windows mappings are case-sensitive.

| Mapping Direction           | Pattern    | Replacement |
|-----------------------------|------------|-------------|
| A Windows user to UNIX user | ENG\\johnp | john        |
| Any Windows user to UNIX    | ENG\\(.+)  | \\1         |
| A UNIX user to Windows      | (.+)       | ENG\\1      |

# The Default User

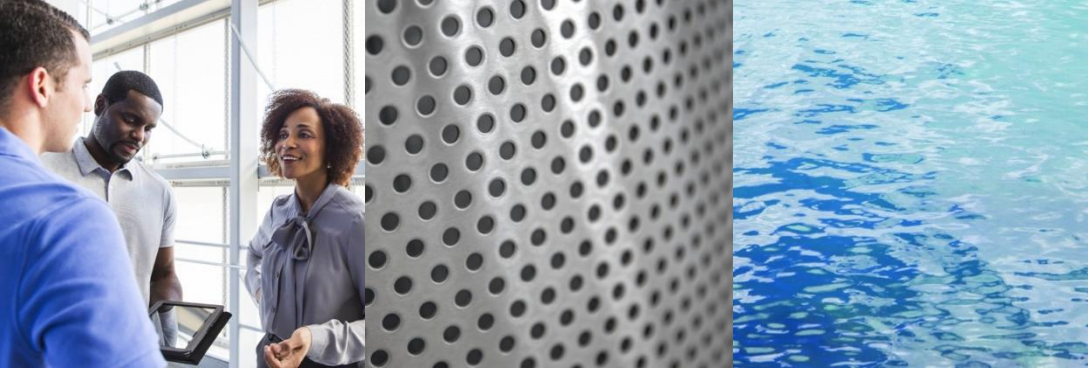
- A Windows user is always mapped to a UNIX user.
- The default user can be used when individual user mappings are not required.
- The default mapping is used if all other mapping attempts fail.
- Authentication fails if no individual mapping or default mapping exists.

Set the default UNIX user:

```
cluster1::>vserver cifs options modify -default-unix-user username
```

Set the default Windows user:

```
cluster1::>vserver nfs modify -default-win-user username
```



## Take a Poll

Multiple-choice Questions

Duration: 5 minutes

# Poll Questions



## 1. Why is multiprotocol needed?

- a. Multiprotocol is required for interoperability between NFS versions 3 and 4.0.
- b. Multiprotocol is required for NFS access by an NFS client.
- c. Multiprotocol allows SMB and NFS clients to access files on the same volume.
- d. There are no scenarios where multiprotocol is needed.

# Poll Questions: Answers



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# Poll Questions



1. Which statement describes the correct use of the default user in a multiprotocol environment?
  - a. Specifying the default user maps all Windows users to a UNIX user with the same name.
  - b. Mapping the default user maps all UNIX users to a single Windows account.
  - c. Mapping the default user maps the root user account to the Windows administrator account.
  - d. Mapping the default user maps the Windows administrator account to the root user account.

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# Poll Questions



2. A file with NTFS permissions has the mixed security style assigned. A UNIX client changes permissions. Which security style is now in effect?
- a. UNIX
  - b. NTFS
  - c. UNIX and NTFS
  - d. Windows

## Poll Questions: Answers



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- d. Windows

# Module Summary



This module focused on enabling you to do the following:

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- Explain the permission mapping between SMB and NFS data



## Complete an Exercise

Module 4: Configuring an SVM for Multiprotocol Access

Duration: 35 minutes

# YOU ARE HERE

- ✓ NFS Fundamentals
- ✓ NFS Version 3
- ✓ NFS Versions 4 and 4.1
- ✓ Multiprotocol Configuration
  - Monitoring and Managing NFS